In [ ]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('anime.csv')

print("Dataset Shape:", df.shape)
display(df.head())

df.info()
```

In [26…
```
df.shape
```

Out[26]:
```
(12294, 7)
```

In [27…
```
df.head()
```

Out[27]:

	anime_id	name	genre	type	episodes	rating	members
0	32281	Kimi no Na wa.	Drama, Romance, School, Supernatural	Movie	1	9.37	200630
1	5114	Fullmetal Alchemist: Brotherhood	Action, Adventure, Drama, Fantasy, Magic, Mili...	TV	64	9.26	793665
2	28977	Gintama°	Action, Comedy, Historical, Parody, Samurai, S...	TV	51	9.25	114262
3	9253	Steins;Gate	Sci-Fi, Thriller	TV	24	9.17	673572
4	9969	Gintama'	Action, Comedy, Historical, Parody, Samurai, S...	TV	51	9.16	151266

In [28…
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 12294 entries, 0 to 12293
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   anime_id  12294 non-null  int64  
 1   name      12294 non-null  str    
 2   genre     12232 non-null  str    
 3   type      12269 non-null  str    
 4   episodes  12294 non-null  str    
 5   rating    12064 non-null  float64
 6   members   12294 non-null  int64  
dtypes: float64(1), int64(2), str(4)
memory usage: 672.5 KB
```

In [29…
```
df.describe()
```

Out[29]:

	anime_id	rating	members
count	12294.000000	12064.000000	1.229400e+04
mean	14058.221653	6.473902	1.807134e+04
std	11455.294701	1.026746	5.482068e+04
min	1.000000	1.670000	5.000000e+00
25%	3484.250000	5.880000	2.250000e+02
50%	10260.500000	6.570000	1.550000e+03
75%	24794.500000	7.180000	9.437000e+03
max	34527.000000	10.000000	1.013917e+06

In [30…
```
df.isnull().sum()
```

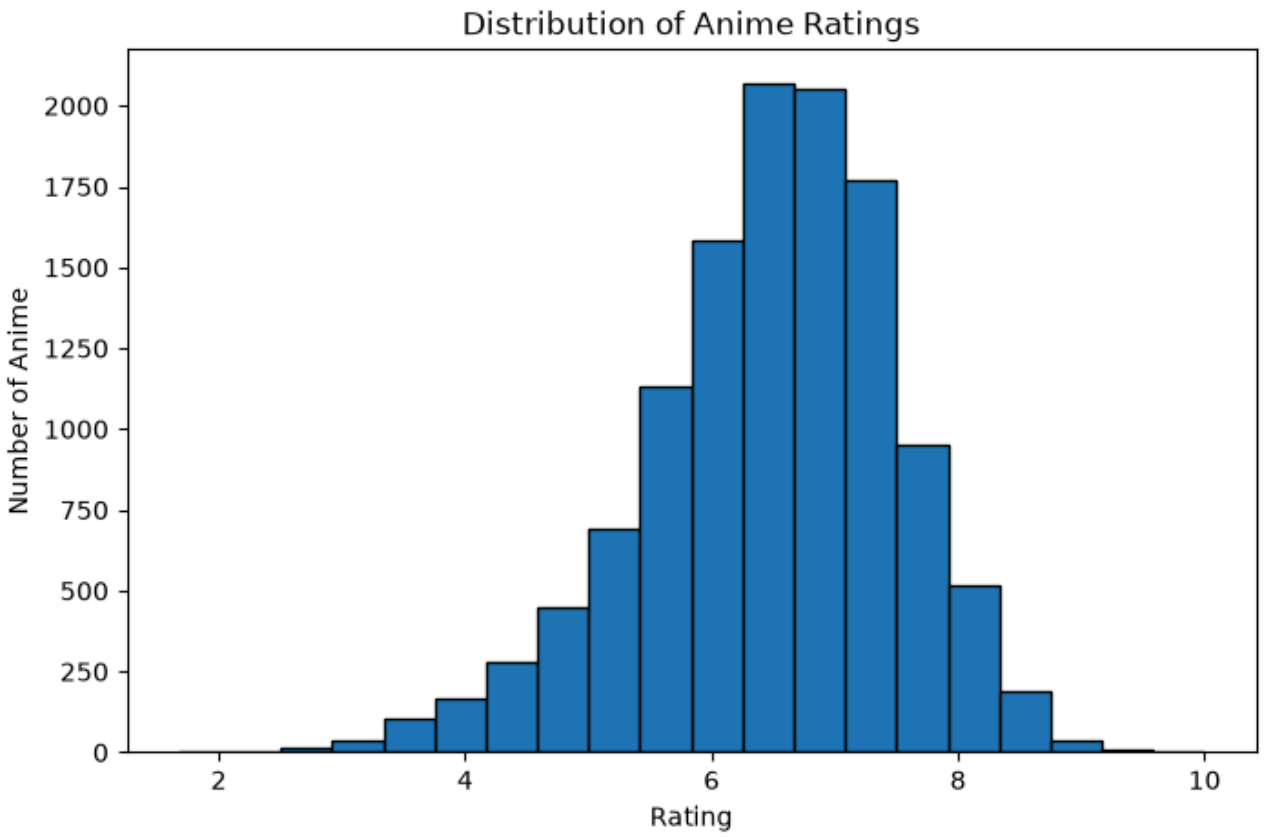
Out[30]:
```
anime_id      0
name          0
genre        62
type         25
episodes      0
rating      230
members       0
dtype: int64
```

In [10…
```
plt.figure(figsize=(8, 5))

plt.hist(df['rating'].dropna(), bins=20, edgecolor='black')

plt.title('Distribution of Anime Ratings')
plt.xlabel('Rating')
plt.ylabel('Number of Anime')

plt.show()
```



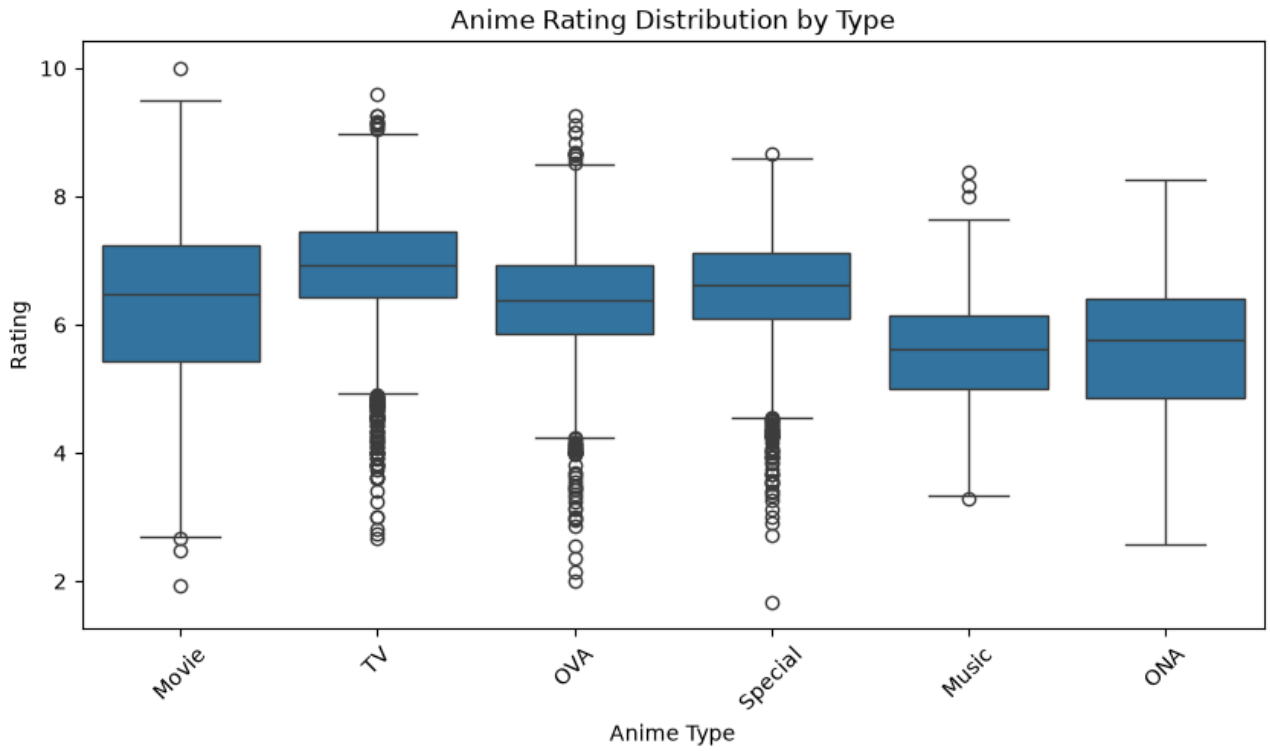
In [11…

```
plt.figure(figsize=(10, 5))

sns.boxplot(data=df, x='type', y='rating')

plt.title('Anime Rating Distribution by Type')
plt.xlabel('Anime Type')
plt.ylabel('Rating')
plt.xticks(rotation=45)

plt.show()
```



In [12…

```
genres = df['genre'].dropna().str.split(', ').explode()

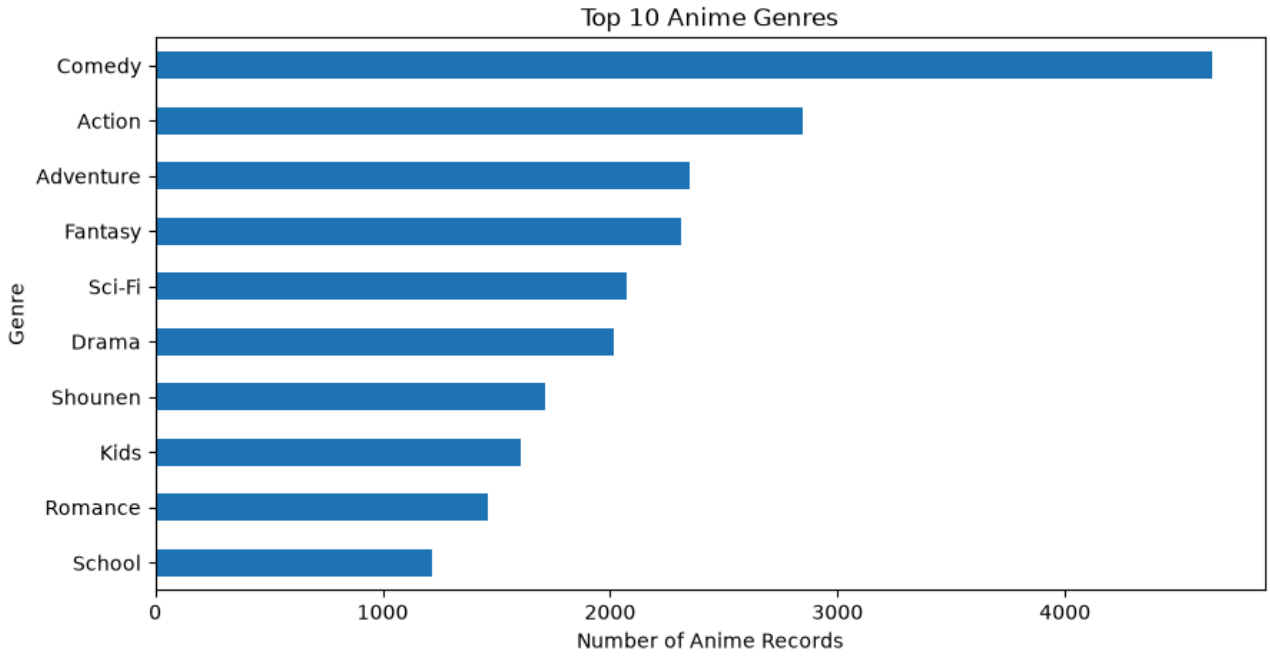
top_genres = genres.value_counts().head(10).sort_values()

plt.figure(figsize=(10, 5))

top_genres.plot(kind='barh')

plt.title('Top 10 Anime Genres')
plt.xlabel('Number of Anime Records')
plt.ylabel('Genre')

plt.show()
```

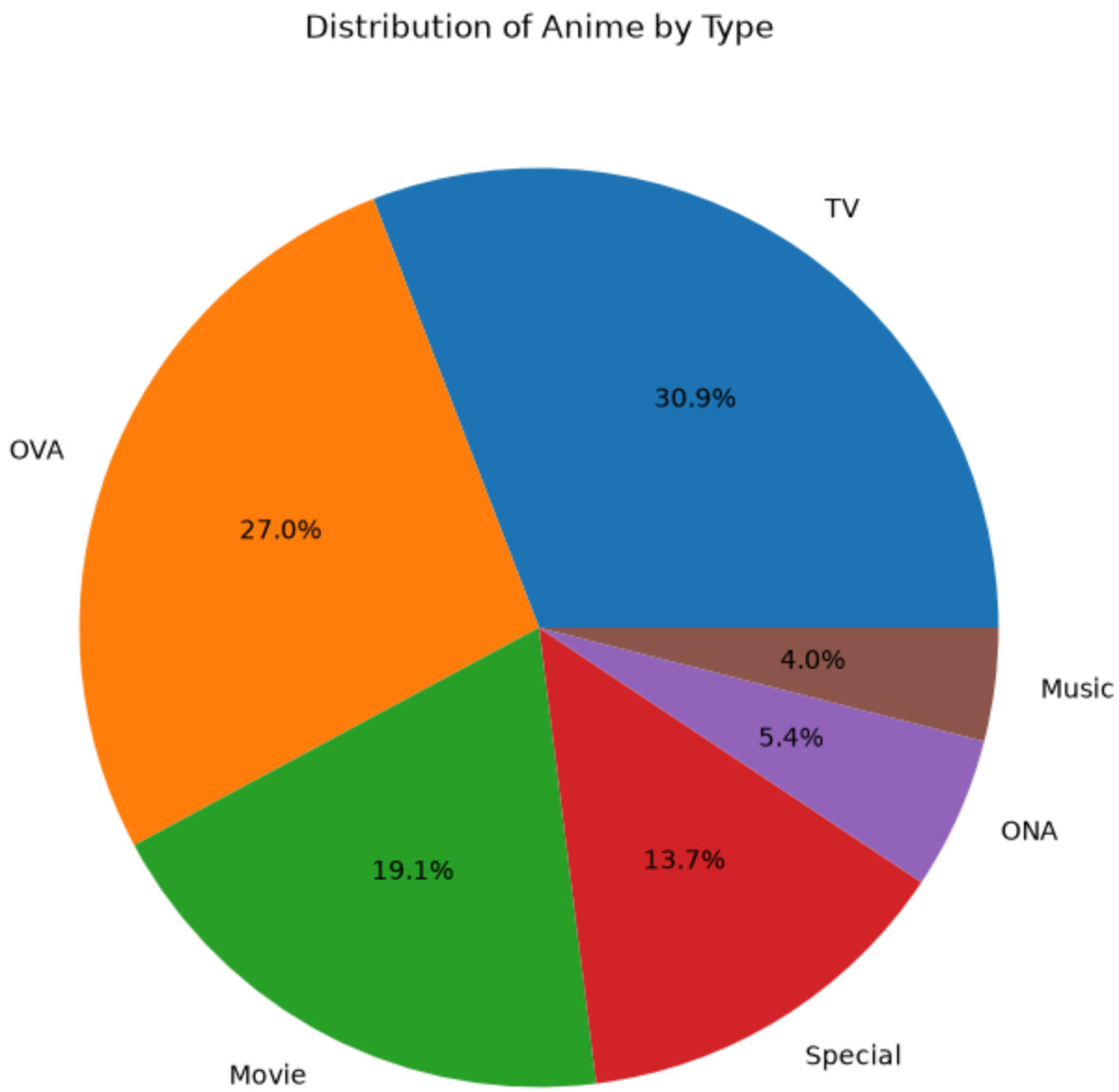


In [13…

```
df['type'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(8, 8)
)

plt.title('Distribution of Anime by Type')
plt.ylabel('')

plt.show()
```



In [14…

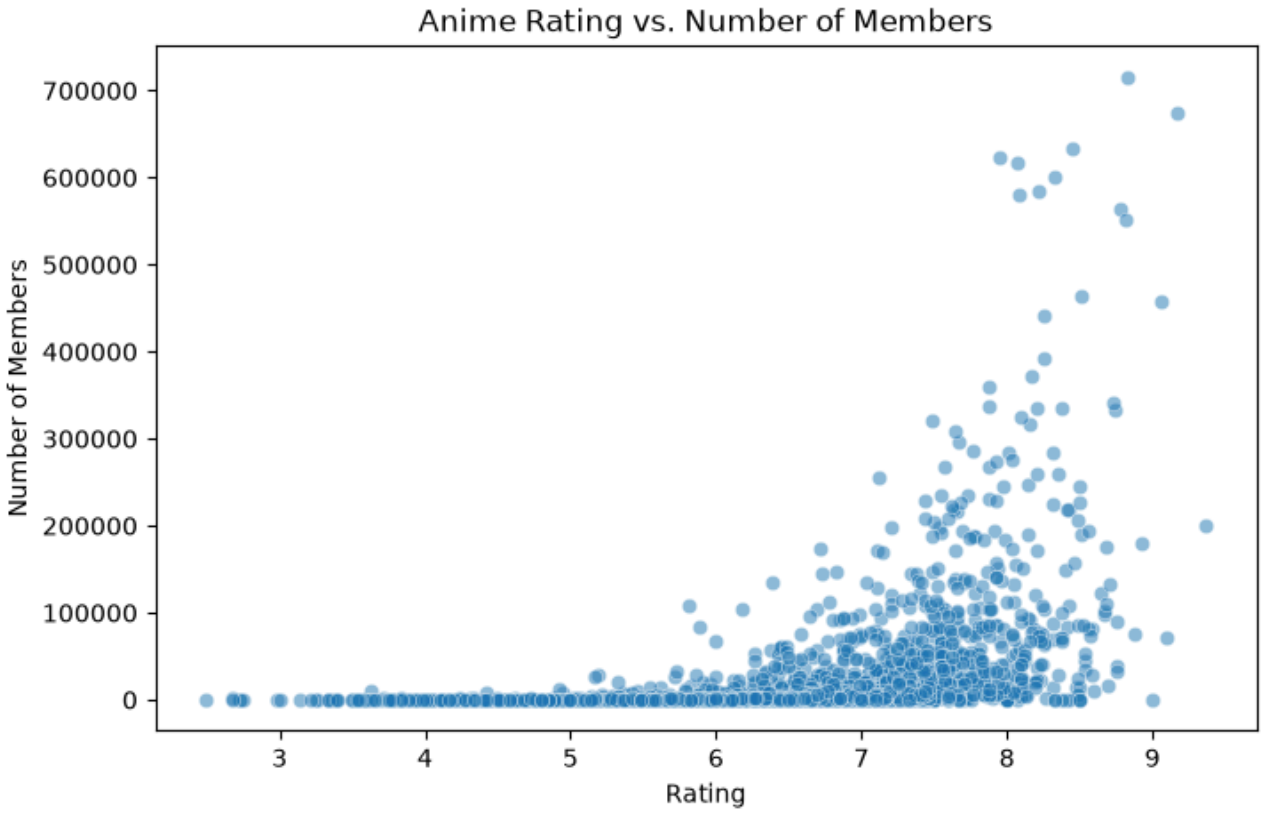
```
sample_df = df.dropna(subset=['rating', 'members']).sample(
    n=min(3000, len(df.dropna(subset=['rating', 'members']))),
    random_state=42
)

plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=sample_df,
    x='rating',
    y='members',
    alpha=0.5
)
```

```
plt.title('Anime Rating vs. Number of Members')
plt.xlabel('Rating')
plt.ylabel('Number of Members')

plt.show()
```



In [15…

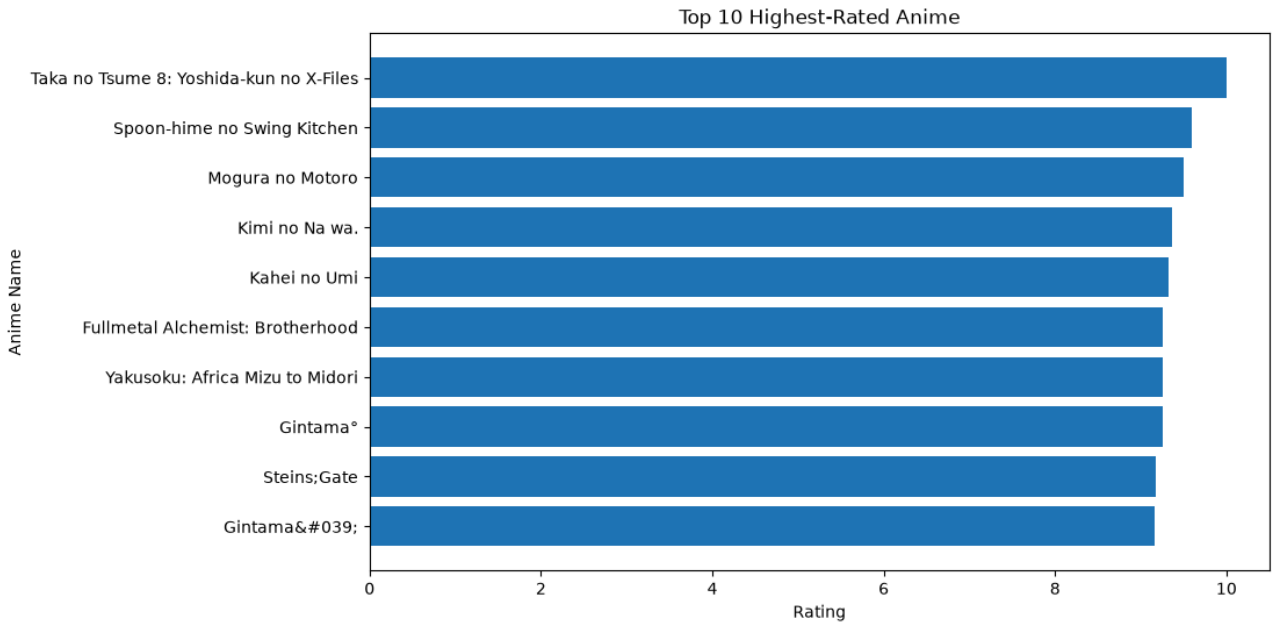
```
top_anime = (
    df.dropna(subset=['rating'])
    .nlargest(10, 'rating')
    .sort_values('rating')
)

plt.figure(figsize=(10, 6))

plt.barh(top_anime['name'], top_anime['rating'])

plt.title('Top 10 Highest-Rated Anime')
plt.xlabel('Rating')
plt.ylabel('Anime Name')

plt.show()
```



In [16…

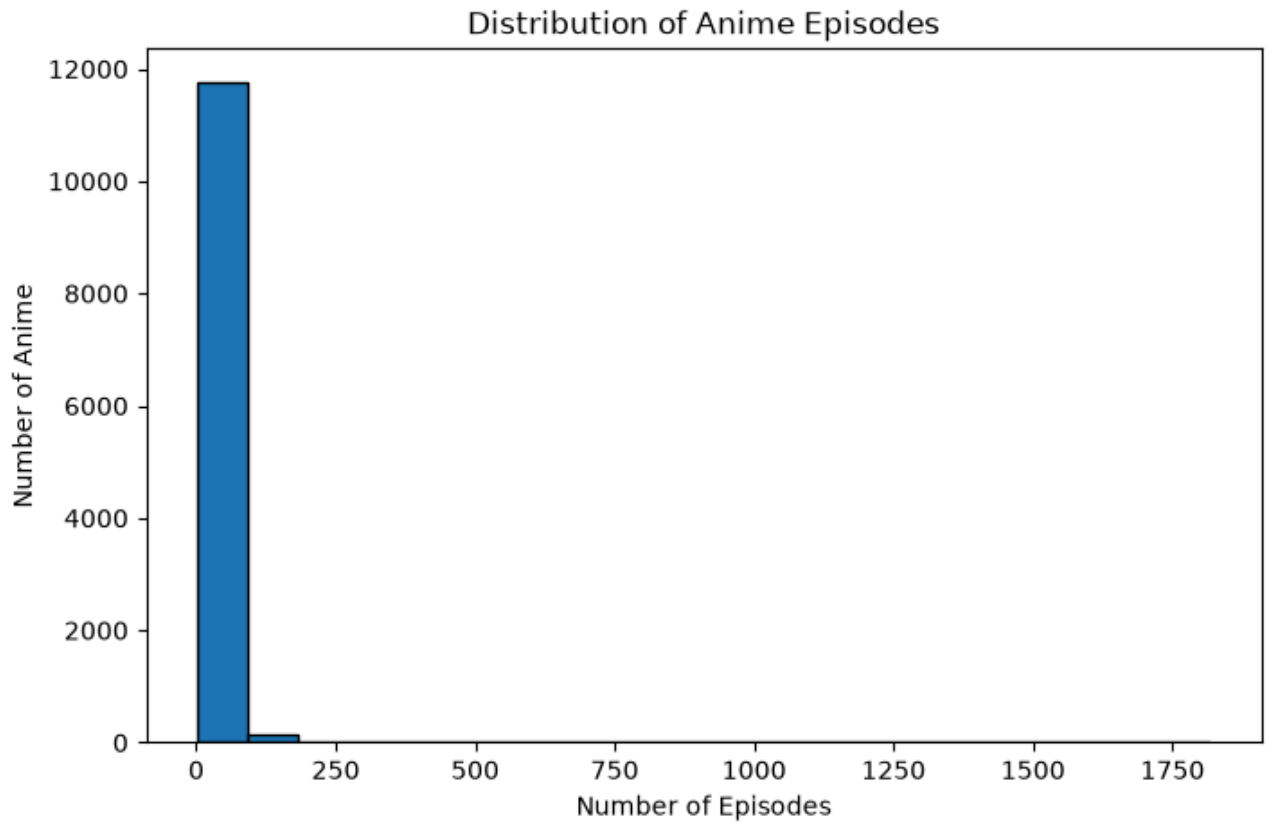
```
df['episodes_numeric'] = pd.to_numeric(
    df['episodes'],
    errors='coerce'
)

plt.figure(figsize=(8, 5))

plt.hist(
    df['episodes_numeric'].dropna(),
    bins=20,
    edgecolor='black'
)

plt.title('Distribution of Anime Episodes')
plt.xlabel('Number of Episodes')
plt.ylabel('Number of Anime')

plt.show()
```



In [17…

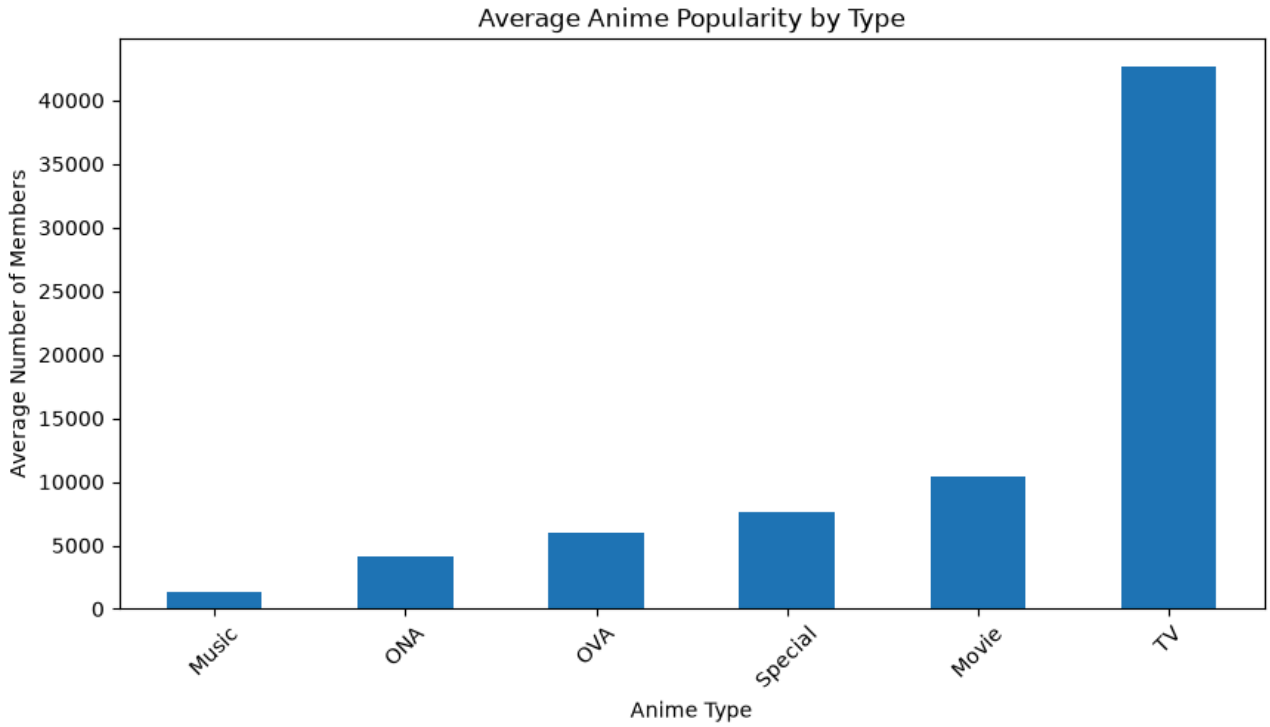
```
popularity = (
    df.groupby('type')['members']
    .mean()
    .sort_values()
)

plt.figure(figsize=(10, 5))

popularity.plot(kind='bar')

plt.title('Average Anime Popularity by Type')
plt.xlabel('Anime Type')
plt.ylabel('Average Number of Members')
plt.xticks(rotation=45)

plt.show()
```



In [18…

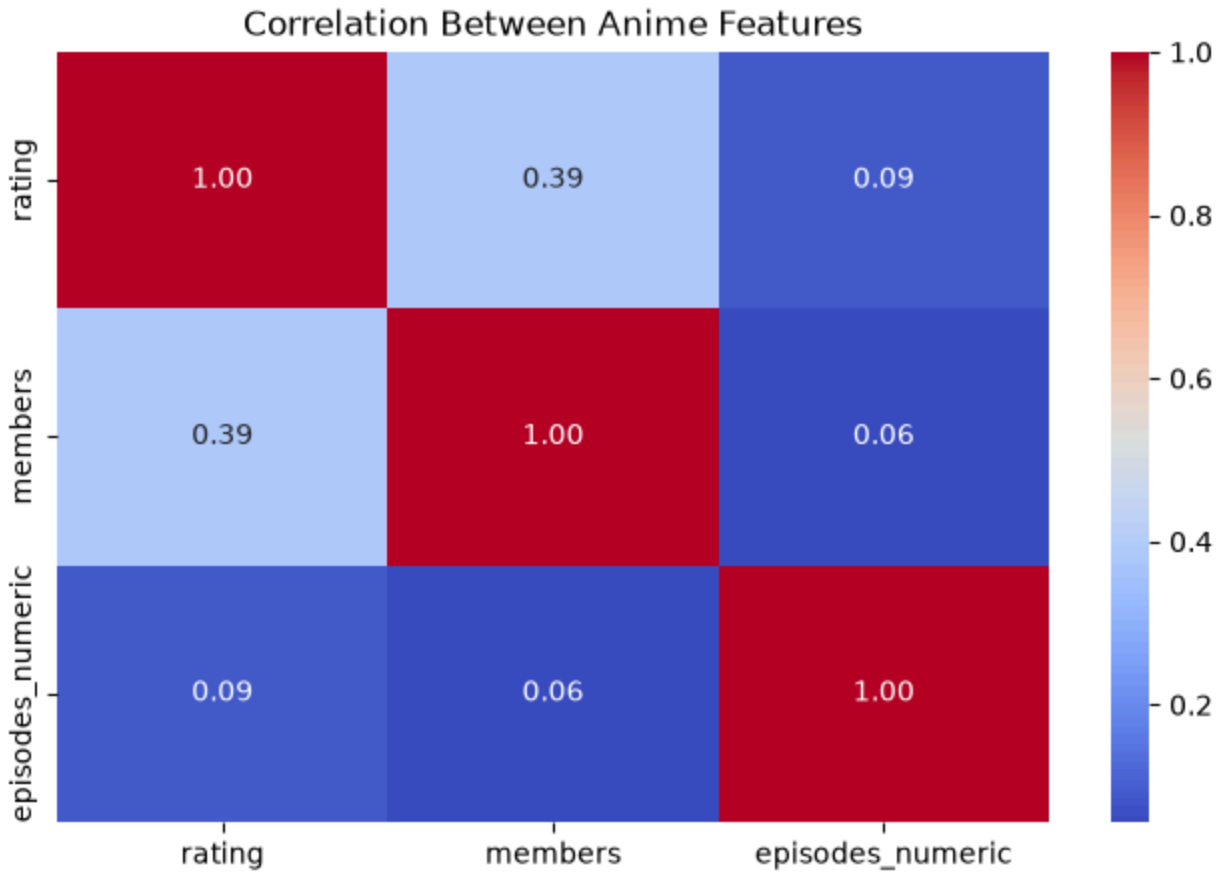
```
numeric_df = df[
    ['rating', 'members', 'episodes_numeric']
]

plt.figure(figsize=(8, 5))

sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap='coolwarm',
    fmt='.2f'
)

plt.title('Correlation Between Anime Features')

plt.show()
```

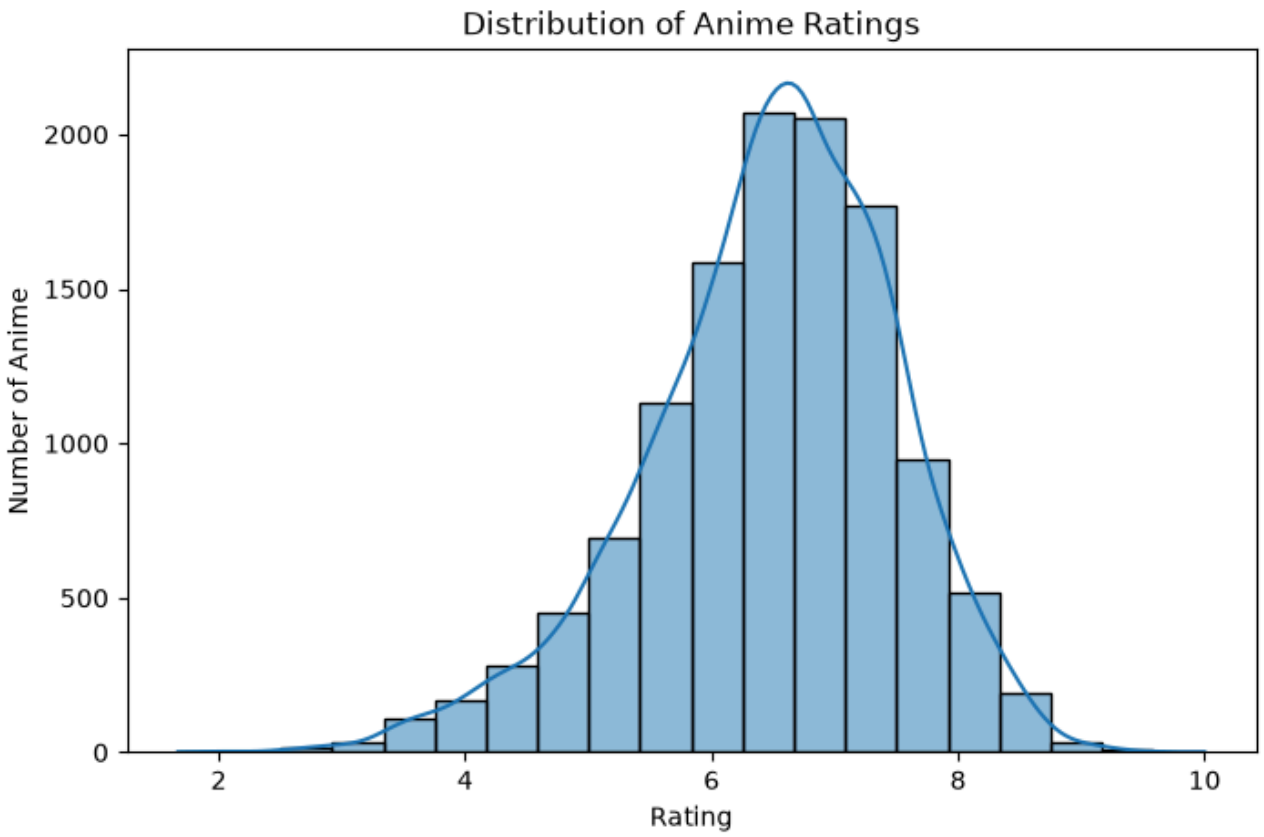


In [19…

```
plt.figure(figsize=(8, 5))

sns.histplot(
    df['rating'].dropna(),
    bins=20,
    kde=True,
    edgecolor='black'
)

plt.title('Distribution of Anime Ratings')
plt.xlabel('Rating')
plt.ylabel('Number of Anime')
plt.show()
```

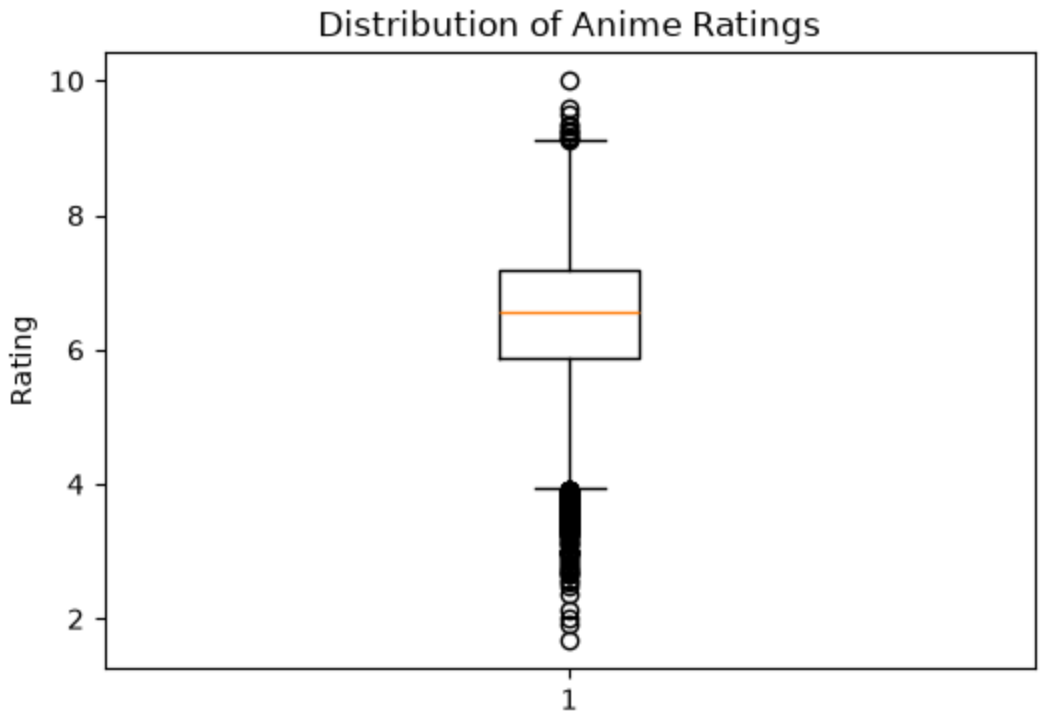


In [20…

```
plt.figure(figsize=(6, 4))

plt.boxplot(df['rating'].dropna())

plt.title('Distribution of Anime Ratings')
plt.ylabel('Rating')
plt.show()
```

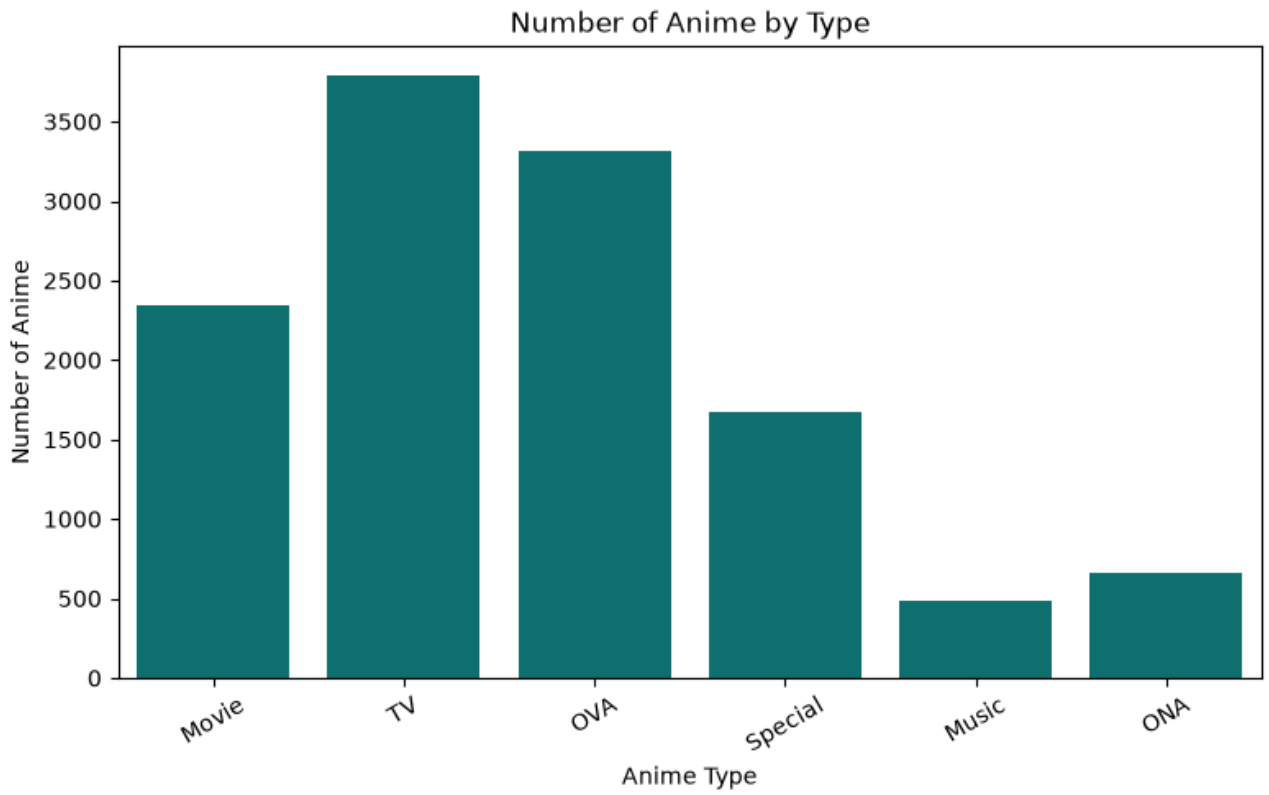


In [21…

```
plt.figure(figsize=(9, 5))

sns.countplot(
    data=df,
    x='type',
    color='teal'
)

plt.title('Number of Anime by Type')
plt.xlabel('Anime Type')
plt.ylabel('Number of Anime')
plt.xticks(rotation=30)
plt.show()
```

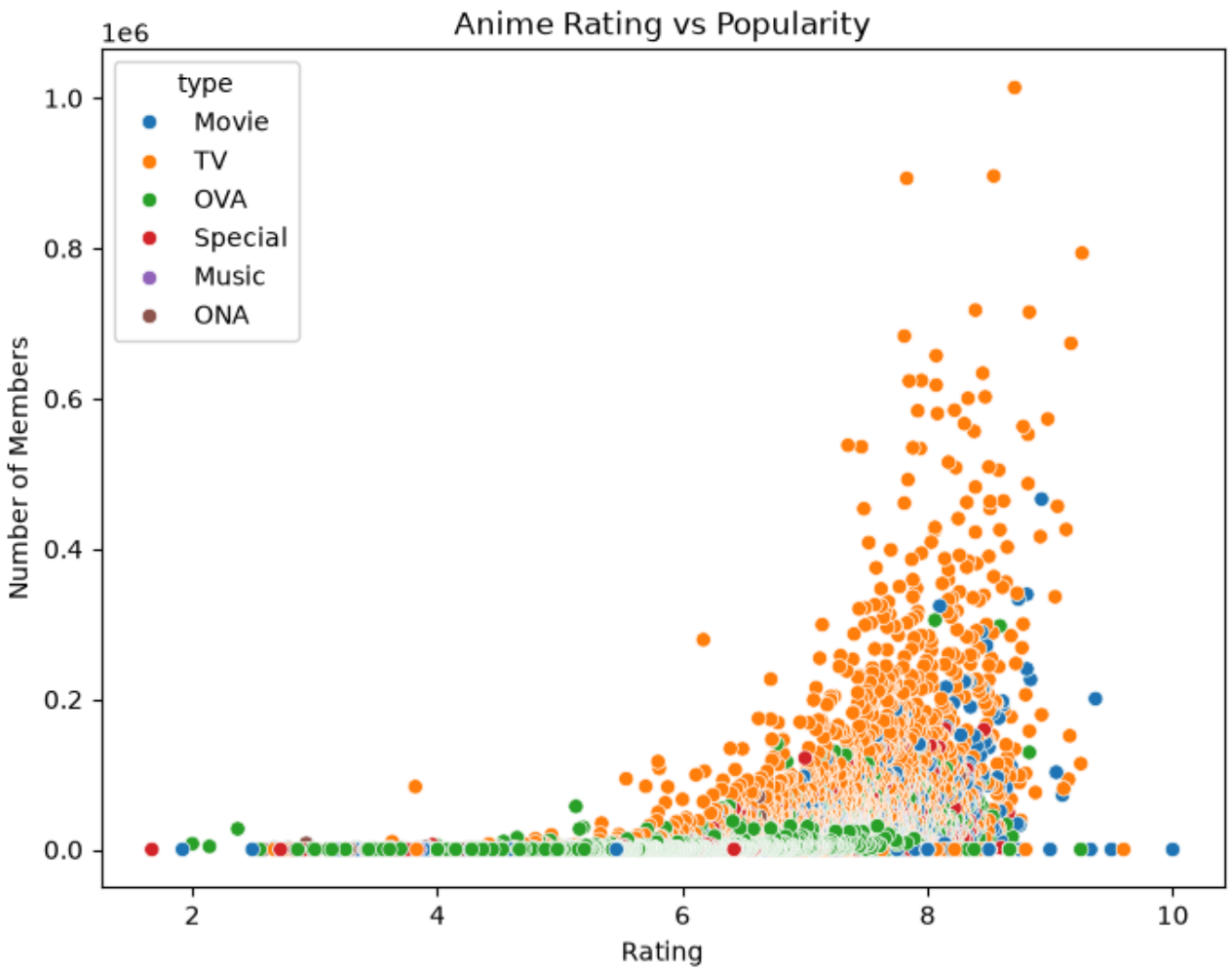


In [22…

```
plt.figure(figsize=(8, 6))

sns.scatterplot(
    data=df,
    x='rating',
    y='members',
    hue='type'
)

plt.title('Anime Rating vs Popularity')
plt.xlabel('Rating')
plt.ylabel('Number of Members')
plt.show()
```

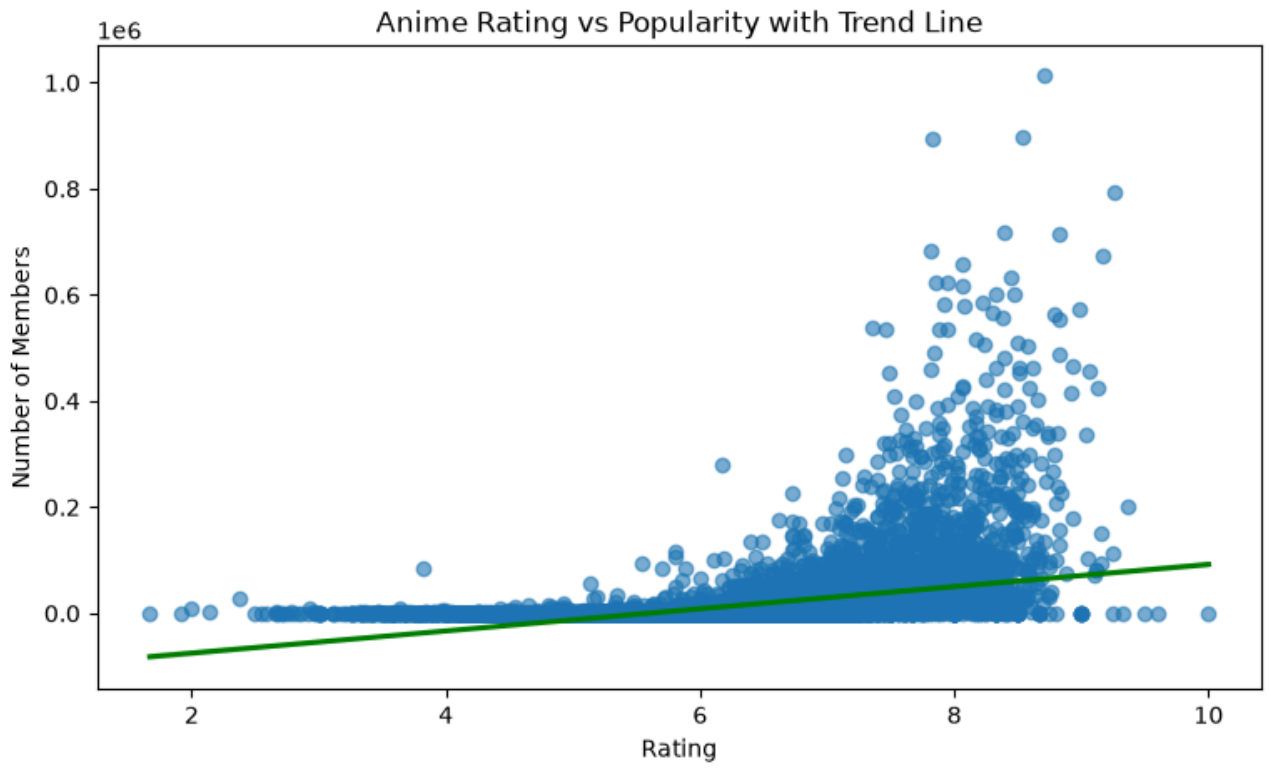


In [23…

```
plt.figure(figsize=(9, 5))

sns.regplot(
    data=df,
    x='rating',
    y='members',
    scatter_kws={'alpha': 0.6},
    line_kws={'color': 'green'}
)

plt.title('Anime Rating vs Popularity with Trend Line')
plt.xlabel('Rating')
plt.ylabel('Number of Members')
plt.show()
```

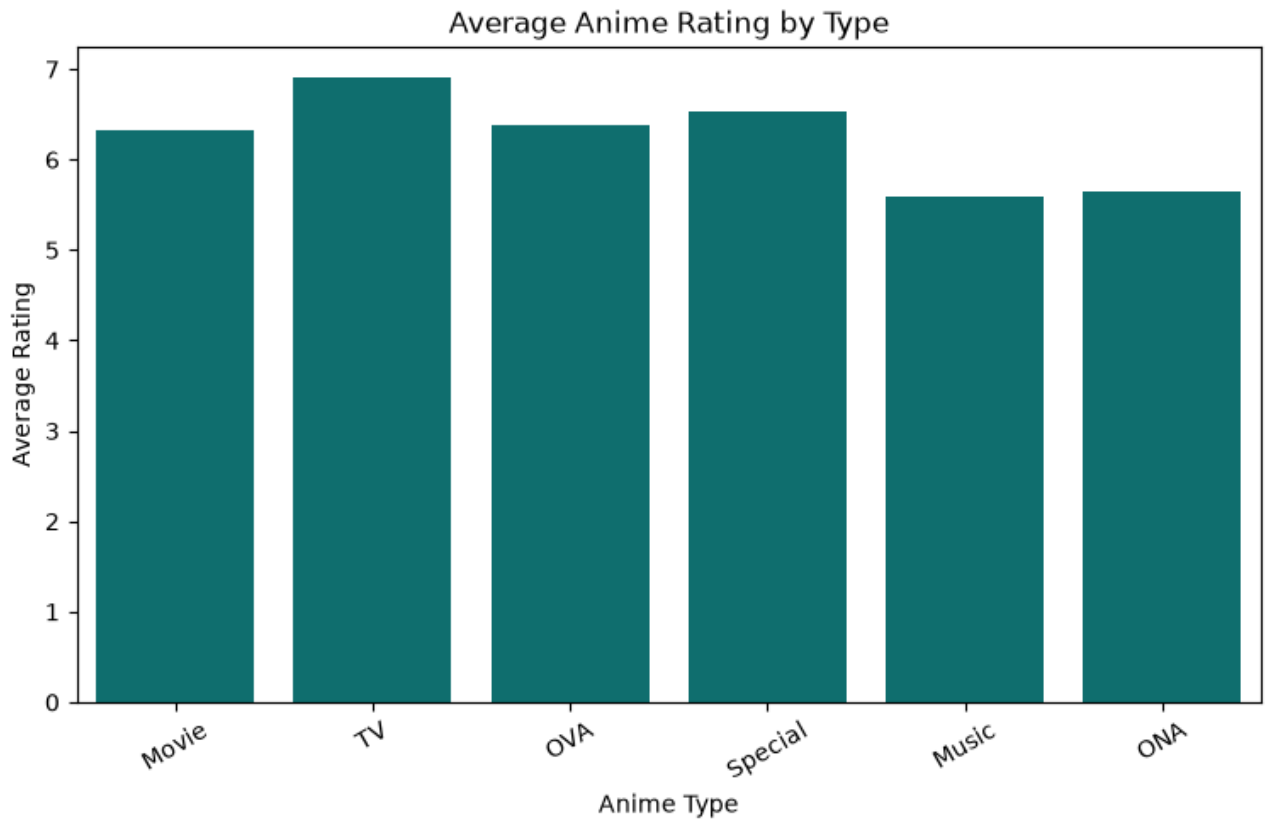


In [24…

```
plt.figure(figsize=(9, 5))

sns.barplot(
    data=df,
    x='type',
    y='rating',
    color='teal',
    errorbar=None
)

plt.title('Average Anime Rating by Type')
plt.xlabel('Anime Type')
plt.ylabel('Average Rating')
plt.xticks(rotation=30)
plt.show()
```


Average Anime Rating by Type
Average Rating
0
1
2
3
4
5
6
7
Movie
TV
OVA
Special
Music
ONA
Anime Type

In [ ]: